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Chromatic index determined by fractional chromatic index

Guantao Chen [a], Yuping Gao [b,a], Ringi Kim [c], Luke Postle [c], Songling Shan [d]

[a] *Department of Mathematics and Statistics, Georgia State University, Atlanta, GA 30303, USA*
[b] *School of Mathematics and Statistics, Lanzhou University, Lanzhou, Gansu 730000, China*
[c] *University of Waterloo, Waterloo, ON, N2L 3G1, Canada*
[d] *Department of Mathematics, Vanderbilt University, Nashville, TN 37240, USA*

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ABSTRACT

Given a graph G possibly with multiple edges but no loops, denote by Δ the maximum degree, μ the multiplicity, χ' the chromatic index and χ'_f the fractional chromatic index of G, respectively. It is known that $\Delta \leq \chi'_f \leq \chi' \leq \Delta + \mu$, where the upper bound is a classic result of Vizing. While deciding the exact value of χ' is a classic NP-complete problem, the computing of χ'_f is in polynomial time. In fact, it is shown that if $\chi'_f > \Delta$ then $\chi'_f = \max \frac{|E(H)|}{\lfloor |V(H)|/2 \rfloor}$, where the maximality is taken over all induced subgraphs H of G. Gupta (1967), Goldberg (1973), Andersen (1977), and Seymour (1979) conjectured that $\chi' = \lceil \chi'_f \rceil$ if $\chi' \geq \Delta + 2$, which is commonly referred as Goldberg's conjecture. It has been shown that Goldberg's conjecture is equivalent to the following conjecture of Jakobsen: For any positive integer m with $m \geq 3$, every graph G with $\chi' > \frac{m}{m-1}\Delta + \frac{m-3}{m-1}$ satisfies $\chi' = \lceil \chi'_f \rceil$. Jakobsen's conjecture has been verified for m up to 15 by various researchers in the last four decades. We use an extended form of a Tashkinov tree to show that it is true for $m \leq 23$. With the same technique, we show that if $\chi' \geq \Delta + \sqrt[3]{\Delta/2}$ then $\chi' = \lceil \chi'_f \rceil$. The previous best known result is for graphs with $\chi' > \Delta + \sqrt{\Delta/2}$ obtained by Scheide, and by Chen, Yu and Zang, independently. Moreover, we show

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that Goldberg's conjecture holds for graphs G with $\Delta \leq 23$ or $|V(G)| \leq 23$.

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1. Introduction

Graphs considered in this paper may contain multiple edges but no loops. Let G be a graph and $\Delta := \Delta(G)$ be the maximum degree of G. A (proper) *k-edge-coloring* φ of G is a mapping φ from $E(G)$ to $\{1, 2, \cdots, k\}$ (whose elements are called colors) such that no two adjacent edges receive the same color. The *chromatic index* $\chi' := \chi'(G)$ is the least integer k such that G has a k-edge-coloring. In graph edge-coloring, the central question is to determine the chromatic index χ' for graphs. We refer to the book [17] of Stiebitz, Scheide, Toft and Favrholdt and the elegant survey [12] of McDonald for literature on the recent progress of graph edge-colorings. Clearly, $\chi' \geq \Delta$. Conversely, Vizing showed that $\chi' \leq \Delta + \mu$, where $\mu := \mu(G)$ is the multiplicity of G. However, determining the exact value of χ' is a very difficult problem. Holyer [8] showed that the problem is NP-hard even restricted to simple cubic graphs. To estimate χ', the notion of fractional chromatic index is introduced.

A *fractional edge coloring* of G is a non-negative weighting $w(.)$ of the set $\mathcal{M}(G)$ of matchings in G such that, for every edge $e \in E(G)$, $\sum_{M \in \mathcal{M}: e \in M} w(M) = 1$. Clearly, such a weighting $w(.)$ exists. The *fractional chromatic index* $\chi'_f := \chi'_f(G)$ is the minimum total weight $\sum_{M \in \mathcal{M}} w(M)$ over all fractional edge colorings of G. By definitions, we have $\chi' \geq \chi'_f \geq \Delta$. It follows from Edmonds' characterization of the matching polytope [3] that χ'_f can be computed in polynomial time and

$$\chi'_f = \max \left\{ \frac{|E(H)|}{\lfloor |V(H)|/2 \rfloor} \ : \ H \subseteq G \text{ with } |V(H)| \geq 3 \right\} \text{ if } \chi'_f > \Delta.$$

It is not difficult to show that the above maximality can be restricted to induced subgraphs H with odd number of vertices. So, in the case of $\chi'_f > \Delta$, we have

$$\lceil \chi'_f \rceil = \max \left\{ \left\lceil \frac{2|E(H)|}{|V(H)| - 1} \right\rceil : \text{induced subgraphs } H \subseteq G \text{ with } |V(H)| \geq 3 \text{ and odd} \right\}.$$

A graph G is called *elementary* if $\chi' = \lceil \chi'_f \rceil$. Gupta (1967) [7], Goldberg (1973) [5], Andersen (1977) [1], and Seymour (1979) [15] independently made the following conjecture, which is commonly referred to as *Goldberg's conjecture*.

Conjecture 1. *For any graph G, if $\chi' \geq \Delta + 2$ then G is elementary.*

An immediate consequence of Conjecture 1 is that χ' can be computed in polynomial time for graphs with $\chi' \geq \Delta + 2$. So the NP-complete problem of computing the chromatic

indices lies in determining whether $\chi' = \Delta$, $\Delta+1$, or $\geq \Delta+2$, which strengthens Vizing's classic result $\chi' \leq \Delta + \mu$ tremendously when μ is big.

Following $\chi' \leq \frac{3\Delta}{2}$ of the classic result of Shannon [16], we can assume that, for every Δ, there exists the least positive number ζ such that if $\chi' > \Delta + \zeta$ then G is elementary. Conjecture 1 indicates that $\zeta \leq 1$. Asymptotically, Kahn [10] showed $\zeta = o(\Delta)$. Scheide [14], and Chen, Yu, and Zang [2] independently proved that $\zeta \leq \sqrt{\Delta/2}$. In this paper, we show that $\zeta \leq \sqrt[3]{\Delta/2} - 1$ as stated below.

Theorem 1.1. *For any graph G, if $\chi' \geq \Delta + \sqrt[3]{\Delta/2}$, then G is elementary.*

Jakobsen [9] conjectured that $\zeta \leq 1 + \frac{\Delta-2}{m-1}$ for every positive integer $m(\geq 3)$, which gives a reformulation of Conjecture 1 as stated below.

Conjecture 2. *Let m be an integer with $m \geq 3$ and G be a graph. If $\chi' > \frac{m}{m-1}\Delta + \frac{m-3}{m-1}$, then G is elementary.*

Since $\frac{m}{m-1}\Delta + \frac{m-3}{m-1}$ decreases as m increases, it is sufficient to prove Jakobsen's conjecture for all odd integers m (in fact, for any infinite sequence of positive integers), which has been confirmed slowly for $m \leq 15$ by a series of papers over the last 40 years:

- $m = 5$**:** Three independent proofs given by Andersen [1] (1977), Goldberg [5] (1973), and Sørensen (unpublished, page 158 in [17]), respectively.
- $m = 7$**:** Two independent proofs given by Andersen [1] (1977) and Sørensen (unpublished, page 158 in [17]), respectively.
- $m = 9$**:** By Goldberg [6] (1984).
- $m = 11$**:** Two independent proofs given by Nishizeki and Kashiwagi [13] (1990) and by Tashkinov [18] (2000), respectively.
- $m = 13$**:** By Favrholdt, Stiebitz and Toft [4] (2006).
- $m = 15$**:** By Scheide [14] (2010).

In this paper, we show that Jakobsen's conjecture is true up to $m = 23$.

Theorem 1.2. *If G is a graph with $\chi' > \frac{23}{22}\Delta + \frac{20}{22}$, then G is elementary.*

Corollary 1.1. *If G is a graph with $\Delta \leq 23$ or $|V(G)| \leq 23$, then $\chi' \leq \max\{\Delta+1, \lceil \chi'_f \rceil\}$.*

Note that in Corollary 1.1, $|V(G)| \leq 23$ does not imply $\Delta \leq 23$, as G may have multiple edges. The remainder of this paper is organized as follows. In Section 2, we introduce some definitions and notation for edge-colorings, Tashkinov trees, and several known results which are useful for the proofs of Theorems 1.1 and 1.2; in Section 3, we give an extension of Tashkinov trees and prove several properties of the extended Tashkinov trees; and in Section 4, we prove Theorem 1.1, Theorem 1.2 and Corollary 1.1 based on the results in Section 3.

2. Preliminaries

2.1. Basic definitions and notation

Let G be a graph with vertex set V and edge set E. Denote by $|G|$ and $||G||$ the number of vertices and the number of edges of G, respectively. For any two sets $X, Y \subseteq V$, denote by $E(X, Y)$ the set of edges with one end in X and the other one in Y and denote by $\partial(X) := E(X, V - X)$ the boundary edge set of X, that is, the set of edges with exactly one end in X. Denote by $G[X]$ the subgraph induced by X and $G - X$ the subgraph induced by $V(G) - X$. Moreover, let $G - x = G - \{x\}$ for every $x \in V(G)$. For any subgraph H of G, we let $G[H] = G[V(H)]$ and $\partial(H) = \partial(V(H))$. Let $V(e)$ be the set of the two ends of an edge e.

A path P is usually denoted by an alternating sequence $P = (v_0, e_1, v_1, \cdots, e_p, v_p)$ with $V(P) = \{v_0, \cdots, v_p\}$ and $E(P) = \{e_1, \cdots, e_p\}$ such that $e_i \in E_G(v_{i-1}, v_i)$ for $1 \leq i \leq p$. For any two vertices $u, v \in V(P)$, denote by uPv or vPu the unique subpath connecting u and v. If u is an end of P, then we obtain a *linear order* $\preceq_{(u,P)}$ of the vertices of P in a natural way such that $x \preceq_{(u,P)} y$ if $x \in V(uPy)$.

The set of all k-edge-colorings of a graph G is denoted by $\mathcal{C}^k(G)$. Let $\varphi \in \mathcal{C}^k(G)$. For any color α, let $E_\alpha^\varphi = \{e \in E \ : \ \varphi(e) = \alpha\}$. More generally, for each subgraph $H \subseteq G$, let

$$E_\alpha^\varphi(H) = \{e \in E(H) \ : \ \varphi(e) = \alpha\}.$$

For any two distinct colors α and β, denote by $G_\varphi(\alpha, \beta)$ the subgraph of G induced by $E_\alpha^\varphi \cup E_\beta^\varphi$. The components of $G_\varphi(\alpha, \beta)$ are called (α, β)-*chains*. Clearly, each (α, β)-chain is either a path or a cycle of edges alternately colored with α and β. For each (α, β)-chain P, let φ/P denote the k-edge-coloring obtained from φ by exchanging colors α and β on P, that is, for each $e \in E$,

$$\varphi/P\ (e) = \begin{cases} \varphi(e), & e \notin E(P); \\ \beta, & e \in E(P) \text{ and } \varphi(e) = \alpha; \\ \alpha, & e \in E(P) \text{ and } \varphi(e) = \beta. \end{cases}$$

For any $v \in V$, let $P_v(\alpha, \beta, \varphi)$ denote the unique (α, β)-chain containing v. Notice that, for any two vertices u, $v \in V$, either $P_u(\alpha, \beta, \varphi) = P_v(\alpha, \beta, \varphi)$ or $P_u(\alpha, \beta, \varphi) \cap P_v(\alpha, \beta, \varphi) = \emptyset$. For any $v \in V$, let $\varphi(v) := \{\varphi(e) \ : \ e \in E(v)\}$ denote the set of colors present at v and $\overline{\varphi}(v)$ the set of colors not assigned to any edge incident to v, which are called *missing* colors at v. For any vertex set $X \subseteq V$, let $\varphi(X) = \cup_{x \in X} \varphi(x)$ and $\overline{\varphi}(X) = \cup_{x \in X} \overline{\varphi}(x)$ be the set of colors present and missing at some vertices of X, respectively. For any edge set $F \subseteq E$, let $\varphi(F) = \cup_{e \in F} \varphi(e)$.

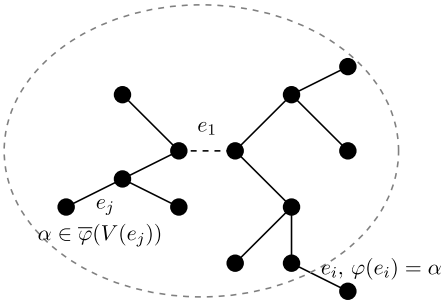

Fig. 1. The construction of a Tashkinov tree; where the dashed curve indicates the boundary of $T[\{e_1, \cdots, e_{i-1}\}]$.

2.2. Elementary sets and closed sets

Let G be a graph. An edge $e \in E(G)$ is called *critical* if $\chi'(G-e) < \chi'(G)$, and the graph G is called *critical* if $\chi'(H) < \chi'(G)$ for any proper subgraph $H \subseteq G$. A graph G is called *k-critical* if it is critical and $\chi'(G) = k+1$. In the proofs, we will consider a graph G with $\chi'(G) = k+1 \geq \Delta + 2$, a critical edge $e \in E(G)$, and a coloring $\varphi \in \mathcal{C}^k(G-e)$. We call them together a *k-triple* (G, e, φ).

Definition 1. Let G be a graph and $e \in E(G)$ such that $\mathcal{C}^k(G-e) \neq \emptyset$ and let $\varphi \in \mathcal{C}^k(G-e)$. Let $X \subseteq V(G)$ contain both ends of e.

- We call X *elementary* (with respect to φ) if all missing color sets $\overline{\varphi}(x)$ $(x \in X)$ are mutually disjoint.
- We call X *closed* (with respect to φ) if $\varphi(\partial(X)) \cap \overline{\varphi}(X) = \emptyset$, i.e., no missing color of X appears on the edges in $\partial(X)$. If additionally, each color in $\varphi(X)$ appears at most once in $\partial(X)$, we call X *strongly closed* (with respect to φ).

Moreover, we call a subgraph $H \subseteq G$ *elementary*, *closed*, and *strongly closed* if $V(H)$ is elementary, closed, and strongly closed, respectively. If a vertex set $X \subseteq V(G)$ containing both ends of e is both elementary and strongly closed, then $|X|$ is odd and $k = \frac{2(|E(G[X])|-1)}{|X|-1}$, so $k+1 = \left\lceil \frac{2|E(G[X])|}{|X|-1} \right\rceil = \lceil \chi'_f \rceil$. Therefore, if $V(G)$ is elementary then G is elementary, i.e., $\chi'(G) = k+1 = \lceil \chi'_f \rceil$.

2.3. Tashkinov trees

Definition 2. A *Tashkinov tree* of a k-triple (G, e, φ) is a tree T, denoted by $T = (e_1, e_2, \cdots, e_p)$, induced by a sequence of edges $e_1 = e, e_2, \ldots, e_p$ such that for each $i \geq 2$, e_i is a boundary edge of the tree induced by $\{e_1, e_2, \cdots, e_{i-1}\}$ and $\varphi(e_i) \in \overline{\varphi}(V(e_j))$ for some $j < i$. (See Fig. 1 for an illustration.)

For each $e_j \in \{e_1, \cdots, e_p\}$, we denote by Te_j the subtree $T[\{e_1, \cdots, e_j\}]$ and denote by e_jT the subgraph of T induced by $\{e_j, \cdots, e_p\}$. For each edge e_i with $i \geq 2$, the

end of e_i in Te_{i-1} is called the *in-end* of e_i and the other one is called the *out-end* of e_i.

Algorithmically, a Tashkinov tree is obtained incrementally from e by adding a boundary edge whose color is missing in the previous tree. Vizing-fans (stars) (used in the proof of Vizing's classic theorem [19]) and Kierstead-paths (used in [11]) are special Tashkinov trees.

Theorem 2.1. (*Tashkinov* [18]) *For any given k-triple (G, e, φ) with $k \geq \Delta+1$, all Tashkinov trees are elementary.*

For a graph G, a Tashkinov tree is associated with an edge $e \in E(G)$ and a k-edge-coloring of $G - e$ with $k \geq \Delta + 1$. We distinguish the following three different types of maximality.

Definition 3. Let (G, e, φ) be a k-triple with $k \geq \Delta + 1$, and T be a Tashkinov tree of (G, e, φ).

- We call T (e, φ)-*maximal* if there is no Tashkinov tree T^* of (G, e, φ) containing T as a proper subtree, and denote by $\mathcal{T}_{e,\varphi}$ the set of all (e, φ)-maximal Tashkinov trees.
- We call T e-*maximal* if there is no Tashkinov tree T^* of a k-triple (G, e, φ^*) containing T as a proper subtree, and denote by $\mathcal{T}_e$ the set of all e-maximal Tashkinov trees.
- We call T *maximum* if $|T|$ is maximum over all Tashkinov trees of G, and denote by $\mathcal{T}$ the set of all maximum Tashkinov trees.

Let T be a Tashkinov tree of a k-triple (G, e, φ). Then, T is (e, φ)-maximal if and only if $V(T)$ is closed. Moreover, the vertex sets are the same for all $T \in \mathcal{T}_{e,\varphi}$. We call colors in $\varphi(E(T))$ *used* and colors not in $\varphi(E(T))$ *unused* on T, call an unused missing color in $\overline{\varphi}(V(T))$ a *free color* of T and denote the set of all free colors of T by $\Gamma^f(T)$. For each color α, let $E_\alpha(\partial(T))$ denote the set of edges with color α in boundary $\partial(T)$. A color α is called a *defective color* of T if $|E_\alpha(\partial(T))| \geq 2$. The set of all defective colors of T is denoted by $\Gamma^d(T)$. Note that if $T \in \mathcal{T}_{e,\varphi}$, then $V(T)$ is strongly closed if and only if T does not have any defective colors.

The following corollary follows immediately from the fact that a maximal Tashkinov tree is elementary and closed.

Corollary 2.1. *For each $T \in \mathcal{T}_{e,\varphi}$, the following properties hold.*

(1) $|T| \geq 3$ *is odd.*
(2) *For any two missing colors $\alpha, \beta \in \overline{\varphi}(V(T))$, we have $P_u(\alpha, \beta, \varphi) = P_v(\alpha, \beta, \varphi)$, where u and v are the two unique vertices in $V(T)$ such that $\alpha \in \overline{\varphi}(u)$ and $\beta \in \overline{\varphi}(v)$, respectively. Furthermore, $V(P_u(\alpha, \beta, \varphi)) \subseteq V(T)$.*

(3) *For every defective color* $\delta \in \Gamma^d(T)$, $|E_\delta(\partial(T))| \geq 3$ *and is odd.*
(4) *There are at least four free colors. More specifically,*

$$|\Gamma^f(T)| \geq |T|(k-\Delta) + 2 - |\varphi(E(T))| \geq |T| + 2 - (|T| - 2) \geq 4.$$

(5) *Let* $T' = (e_1, e_2, \cdots, e_q)$ *be any Tashkinov tree of a k-triple* (G, e, φ). *Then there exists* $\widehat{T} \in \mathcal{T}_{e,\varphi}$ *such that* $V(\widehat{T}) = V(T)$ *and* $\widehat{T}e_q = T'$.

The following lemma was given in [17].

Lemma 2.1. *Let* $T \in \mathcal{T}_e$ *be a Tashkinov tree of a k-triple* (G, e, φ) *with* $k \geq \Delta + 1$. *For any free color* $\gamma \in \Gamma^f(T)$ *and any* $\delta \notin \overline{\varphi}(V(T))$, *the* (γ, δ)*-chain* $P_u(\gamma, \delta, \varphi)$ *contains all edges in* $E_\delta(\partial(T))$, *where u is the unique vertex of T missing color* γ.

Proof. Otherwise, consider the coloring $\varphi_1 = \varphi/P_u(\gamma, \delta, \varphi)$. Since δ and γ are both unused on T with respect to φ, T is still a Tashkinov tree and δ is a missing color with respect to φ_1. But $E^\varphi_\delta(\partial(T)) \neq \emptyset$, which gives a contradiction to T being an e-maximal tree. □

Following the notation in Lemma 2.1, we consider the case of δ being a defective color. Then $P := P_u(\gamma, \delta, \varphi)$ is a path with u as one end. Since u is the unique vertex in T missing γ by Theorem 2.1, the other end of P is not in T. In the linear order $\preceq_{(u,P)}$, the last vertex v with $v \in V(T) \cap V(P)$ is called an *exit vertex* of T. Applying Lemma 2.1, Scheide [14] obtained the following result.

Lemma 2.2. *Let* $T \in \mathcal{T}_e$ *be a Tashkinov tree of a k-triple* (G, e, φ) *with* $k \geq \Delta + 1$. *If v is an exit vertex of T, then every missing color in* $\overline{\varphi}(v)$ *must be used on T.*

For $e \in E(G)$ with $V(e) = \{x, y\}$, by keeping an odd number of vertices in each step of growing a Tashkinov tree from e, we can construct a Tashkinov tree incrementally from e by the following steps:

- **Adding a path:** Pick two missing colors α and β with $\alpha \in \overline{\varphi}(x)$ and $\beta \in \overline{\varphi}(y)$, and let $T^* := \{e\} \cup (P_x(\alpha, \beta, \varphi) - y)$ where $P_x(\alpha, \beta, \varphi)$ is the (α, β)-chain containing both x and y.
- **Adding edges by pairs:** Repeatedly pick two boundary edges f_1 and f_2 of T^* with $\varphi(f_1) = \varphi(f_2) \in \overline{\varphi}(V(T^*))$ and redefine $T^* := T^* \cup \{f_1, f_2\}$ until T^* is closed.

A Tashkinov tree constructed in this way is called a *balanced Tashkinov tree*. The path $P_x(\alpha, \beta, \varphi)$ in the above definition is called the *trunk* of T^* and $h(T^*) := |V(P_x(\alpha, \beta, \varphi))|$ is called the *height* of T^*. A balanced Tashkinov tree is introduced by Scheide [14] and showed that for $T \in \mathcal{T}_{e,\varphi}$, there is a balanced Tashkinov tree $T^* \in \mathcal{T}_{e,\varphi}$ such that $V(T^*) = V(T)$. We also present the idea showing its existence below.

Lemma 2.3. (Scheide [14]) *For $T \in \mathcal{T}_{e,\varphi}$, there exists a balanced Tashkinov tree $T^* \in \mathcal{T}_{e,\varphi}$ such that $V(T^*) = V(T)$.*

Proof. Let $V(e) = \{x, y\}$, $\alpha \in \overline{\varphi}(x)$, and $\beta \in \overline{\varphi}(y)$. Since e is critical, we know that $P_x(\alpha, \beta, \varphi) = P_y(\alpha, \beta, \varphi)$. Thus, $T_1 := \{e\} \cup (P_x(\alpha, \beta, \varphi) - y)$ is a Tashkinov tree. Let e' be the last edge of T_1. By (5) of Corollary 2.1, we know that there exists $\widehat{T} \in \mathcal{T}_{e,\varphi}$ such that $\widehat{T}e' = T_1$ and $V(\widehat{T}) = V(T)$. Let $\widehat{T} = (e_1, e_2, \cdots, e_p)$, and let T' be a balanced Tashkinov tree with respect to e and φ so that $|V(T') \cap V(\widehat{T})|$ is maximum. We may assume that there exists a smallest index i, $1 < i \leq p$, such that $V(\widehat{T}e_{i-1}) \subseteq V(T')$ and $V(\widehat{T}e_i) \nsubseteq V(T')$. By the construction of T', it has odd number of vertices. Since $\varphi(e_i)$ is missing in $\overline{\varphi}(V(T'))$, and T' is elementary by Theorem 2.1, there exists another edge $f \in \partial(T')$ distinct from e_i such that $\varphi(f) = \varphi(e_i)$. Hence, $T' \cup \{e_i, f\}$ is balanced Tashkinov tree which contains more vertices of $\widehat{T}$ than T' does, showing a contradiction to the choice of T'. So there exists a balanced Tashkinov tree $T^* \in \mathcal{T}_{e,\varphi}$ such that $V(T) = V(\widehat{T}) \subseteq V(T^*)$. Since T is (e, φ)-maximal, we conclude that $V(T^*) = V(\widehat{T}) = V(T)$. □

In applications, balanced Tashkinov trees are more useful. As stressed in the lemma below, its construction guarantees that at least half of the missing colors of the tree are actually free colors. And the existence of many free colors gives more room in performing Kempe swap with one color taken as a free color and the other taken as an arbitrary missing color of the balanced Tashkinov tree, thereby gives us more power later on in the proof.

Lemma 2.4. (Scheide [14]) *Let G be a k-critical graph with $k \geq \Delta + 1$ and $T \in \mathcal{T}$ be a balanced Tashkinov tree of a k-triple (G, e, φ) with $h(T)$ being maximum. Then, $h(T) \geq 3$ is odd. Moreover, if $h(T) = 3$ then G is elementary.*

Corollary 2.2. *Let G be a non-elementary k-critical graph with $k \geq \Delta + 1$ and $T \in \mathcal{T}$ be a balanced Tashkinov tree of a k-triple (G, e, φ) with $h(T)$ being maximum. Then $|T| \geq 2(k - \Delta) + 1$.*

Proof. Since G is not elementary, T is not strongly closed with respect to φ. There is an exit vertex v by Lemma 2.1, so $\overline{\varphi}(v) \subseteq \varphi(E(T))$ by Lemma 2.2. Since T is balanced and $h(T) \geq 5$ by Lemma 2.4, each used color is assigned to at least two edges of $E(T)$. Thus,

$$|T| = ||T|| + 1 \geq 2|\overline{\varphi}(v)| + 1 \geq 2(k - \Delta) + 1. \quad \square$$

Working on balanced Tashkinov trees, Scheide proved the following result.

Lemma 2.5. (Scheide [14]) *Let G be a k-critical graph with $k \geq \Delta + 1$. If $|T| < 11$ for all Tashkinov trees T, then G is elementary.*

3. An extension of Tashkinov trees

3.1. Definitions and basic properties

In this section, we always assume that G is a *non-elementary* k-critical graph with $k \geq \Delta + 1$ and fix a maximum Tashkinov tree $T_0 \in \mathcal{T}$ of G. Let e be the first element of T_0 and Φ the set of all colorings $\varphi \in \mathcal{C}^k(G - e)$ such that T_0 is a Tashkinov tree of (G, e, φ).

Definition 4. Let $\varphi_1, \varphi_2 \in \mathcal{C}^k(G - e)$ and $H \subseteq G$ such that $e \in E(H)$. We say that H is (φ_1, φ_2)-*stable* if $\varphi_1(f) = \varphi_2(f)$ for every $f \in E(G[V(H)]) \cup \partial(H)$, that is, $\varphi_1(f) \neq \varphi_2(f)$ implies that $f \in E(G - V(H))$.

Following the definition, if a Tashkinov tree T_0 of (G, e, φ_1) is (φ_1, φ_2)-stable, then it is also a Tashkinov tree of (G, e, φ_2). Moreover, the sets of missing colors of T_0, used colors of T_0, and free colors of T_0 are the same in both colorings φ_1 and φ_2. Thus, for $\varphi_1, \varphi_2 \in \mathcal{C}^k(G - e)$, if T_0 is (φ_1, φ_2)-stable then $\varphi_1 \in \Phi$ if and only if $\varphi_2 \in \Phi$.

The following definition of *connecting edges* will play a critical role in our extension based on a maximum Tashkinov tree.

Definition 5. Let $H \subseteq G$ be a subgraph such that $T_0 \subseteq H$ and $\varphi \in \Phi$. A color δ is called a *defective color* of H (with respect to φ) if H is closed, $\delta \notin \overline{\varphi}(V(H))$ and $|E_\delta^\varphi(\partial(H))| \geq 2$. Moreover, an edge $f \in \partial(H)$ is called a *connecting edge* if $\delta := \varphi(f)$ is a defective color of H and there is a missing color $\gamma \in \overline{\varphi}(V(T_0)) - \varphi(E(H))$ of T_0 such that the following two properties hold.

- The (γ, δ)-chain $P_u(\delta, \gamma, \varphi)$ contains all edges in $E_\delta^\varphi(\partial(H))$, where u is the unique vertex in $V(T_0)$ such that $\gamma \in \overline{\varphi}(u)$;
- Along the linear order $\preceq_{(u, P_u(\gamma, \delta, \varphi))}$, f is the first boundary edge on $P_u(\gamma, \delta, \varphi)$ with color δ.

In the above definition, we call the successor f^s of f along $\preceq_{(u, P_u(\gamma, \delta, \varphi))}$ the *companion* of f, (f, f^s) a *connecting edge pair* and (δ, γ) a *connecting color pair*. Since $P_u(\gamma, \delta, \varphi)$ contains all edges in $E_\delta^\varphi(\partial(H))$, we have that f^s is not incident to any vertex in H and $\varphi(f^s) = \gamma$.

For two sequences $X = (x_1, x_2, \ldots, x_n)$ and $Y = (y_1, y_2, \ldots, y_m)$, let $X + Y$ be the sequence $(x_1, x_2, \ldots, x_n, y_1, \ldots, y_m)$. Recall that a Tashkinov tree is indeed a sequence of edges.

Definition 6. An **Extension of a Tashkinov Tree (ETT)** of (G, e) based on T_0 is a pair (T, φ) of a sequence $T = T_0 + (e_1, \ldots, e_n)$ of edges of G and $\varphi \in \Phi$ such that T is incrementally obtained from $T := T_0$ by repeatedly adding edges to T according to the following two operations subject to $\Gamma^f(T_0) - \varphi(E(T)) \neq \emptyset$:

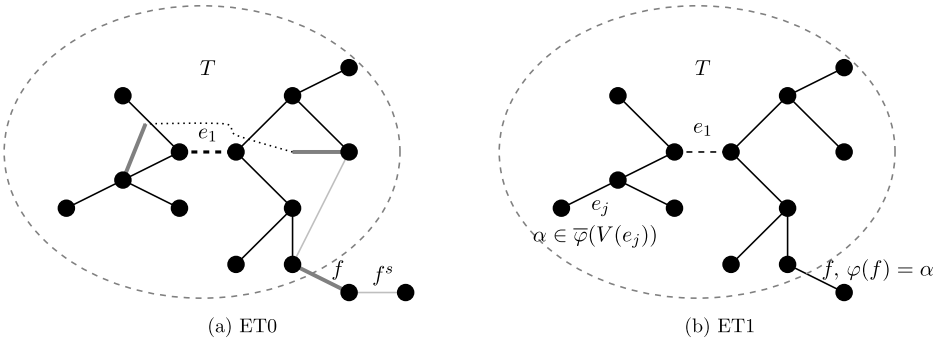

Fig. 2. The construction of an ETT; in (a), the thick gray and gray alternating path is the path containing the edges f, f_s.

- **ET0:** If T is closed, add a connecting edge pair (f, f^s), where $\varphi(f)$ is a defective color and $\varphi(f^s) \in \Gamma^f(T_0) - \varphi(E(T))$, and rename $T := T \cup (f, f^s)$.
- **ET1:** Otherwise, add an edge $f \in \partial(T)$ with $\varphi(f) \in \overline{\varphi}(V(T))$ being a missing color of T, and rename $T := T \cup \{f\}$.

We denote by $E(T)$ the set $E(T_0) \cup \{e_1, e_2, \ldots, e_n\}$ and $V(T)$ the vertex set of the subgraph of G induced by $E(T)$. Fig. 2 demonstrates the constructions of an ETT.

Note that the above extension algorithm ends with $\Gamma^f(T_0) \subseteq \varphi(E(T))$. Of course, an ETT can be defined for every maximum Tashkinov tree, but in this section, we only consider ETTs based on T_0.

Let (T, φ) be an ETT. For $e, f \in E(T)$, we denote by $e \prec_T f$ if e comes before f in T. For any initial subsequence S of T containing T_0, (S, φ) induces an ETT; we call it a *premier segment* of (T, φ) provided that when a connecting edge is in S, its companion must be in S. Let $f_1, f_2, \ldots, f_{m+1}$ be all connecting edges with $f_1 \prec_T f_2 \prec_T \cdots \prec_T f_{m+1}$. For each $1 \le i \le m+1$, let T_{i-1} be the initial subsequence of edges of T coming before f_i. Clearly, we have $T_0 \subset T_1 \subset T_2 \subset \cdots \subset T_m \subset T$. We call (T_i, φ) a *closed segment* of (T, φ) for each $0 \le i \le m$, $T_0 \subset T_1 \subset T_2 \subset \cdots \subset T_m \subset T$ the *ladder of* (T, φ), and (T, φ) an *ETT with m rungs*. We use $m(T, \varphi)$ to denote the number of rungs of T. For each edge $f \in E(T)$ with $f \neq e$, following the linear order $\prec_T$, the end of f is called the *in-end* if it is in T before f and the other one is called the *out-end* of f. See Fig. 3 for a display of the ladder structure of T.

For ETTs $(T, \varphi), (T', \varphi')$, we say (T, φ) *is equivalent to* (T', φ'), write $(T, \varphi) \sim (T', \varphi')$, if $T = T'$ and $f \in T$ is a connecting edge of (T, φ) if and only if f is a connecting edge of (T', φ'). So, if $(T, \varphi) \sim (T', \varphi')$, then the constructions of (T, φ) and (T', φ') are the same. Also note that (T, φ) is equivalent to (T', φ') if and only if $T = T'$ and the ladders of (T, φ) and (T', φ') are the same.

For an ETT (T, φ) with ladder $T_0 \subset T_1 \subset \cdots \subset T_m \subset T$, we say (T, φ) is maximal if for every closed segment (T_i, φ) of (T, φ), T_i is closed with respect to φ' for every $\varphi' \in \Phi$ such that (T_i, φ') is an ETT equivalent to (T_i, φ).

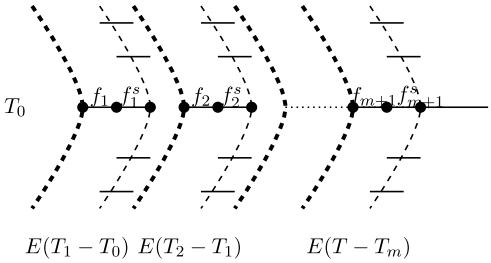

Fig. 3. The ladder structure of an ETT; where thick dashed curves indicate the boundaries of T_i's, and the thin dashed curves indicate the boundaries of the trees after each time adding connecting edge pairs f, f_s, and the edges between two consecutive thick dashed curves are the edges added to T_{i-1} to get T_i.

By the definition of a maximal ETT, the following lemma is straightforward since every closed segment of a premier segment of an ETT (T, φ) is a closed segment of (T, φ).

Lemma 3.1. *If (T, φ) is a maximal ETT, then every premier segment (T', φ) of T is a maximal ETT.*

Lemma 3.2. *Let (T, φ_1) be a maximal ETT, and let $T_0 \subset T_1 \subset \cdots \subset T_m \subset T$ be the ladder of (T, φ_1). Then for every $\varphi_2 \in \Phi$ such that T_m is (φ_1, φ_2)-stable, (T_m, φ_2) is an ETT equivalent to (T_m, φ_1). Furthermore, for $\gamma \in \Gamma^f(T_0) - \varphi_2(E(T_m))$ and $\delta \notin \overline{\varphi}_2(V(T_m))$, if $P_u(\gamma, \delta, \varphi_2) \cap E_\delta^{\varphi_2}(\partial(T_m)) \neq \emptyset$ then $P_u(\gamma, \delta, \varphi_2) \supseteq E_\delta^{\varphi_2}(\partial(T_m))$ where $u \in V(T_0)$ such that $\gamma \in \overline{\varphi}_2(u)$.*

Proof. Suppose on the contrary: let T be a counterexample to Lemma 3.2 with minimum number of vertices. Let $\varphi_1, \varphi_2 \in \Phi$ be two edge colorings such that (T, φ_1) is an ETT with the ladder $T_0 \subset \cdots \subset T_m \subset T$, T_m is (φ_1, φ_2)-stable and either

(1) (T_m, φ_2) is not an ETT equivalent to (T_m, φ_1) or
(2) (T_m, φ_2) is an ETT equivalent to (T_m, φ_1) and there exist $\gamma \in \Gamma^f(T_0) - \varphi_2(E(T_m))$ and $\delta \notin \overline{\varphi}_2(V(T_m))$ such that $P_u(\gamma, \delta, \varphi_2) \cap E_\delta^{\varphi_2}(\partial(T_m)) \neq \emptyset$ and $P_u(\gamma, \delta, \varphi_2) \not\supseteq E_\delta^{\varphi_2}(\partial(T_m))$ where $u \in V(T_0)$ such that $\gamma \in \overline{\varphi}_1(u)(= \overline{\varphi}_2(u))$.

By the minimality of T, we observe that $|T| = |T_m| + 2$. Furthermore, since $T_0 \in \mathcal{T}$ is a maximum Tashkinov tree of G, it follows that $m \geq 1$ by Lemma 2.1.

First, we show that (1) does not hold, in other words, (T_m, φ_2) is an ETT equivalent to (T_m, φ_1). Since colors for edges incident to vertices in T_m are the same in both φ_1 and φ_2, we only need to show that each connecting edge pair in coloring φ_1 is still a connecting edge pair in coloring φ_2. For $0 \leq j \leq m-1$ let (f_j, f_j^s) be the connecting edge pair of T_j in (T, φ_1) and let (δ_j, γ_j) be the corresponding connecting color pair with respect to φ_1. Let $P_j := P_{u_j}(\delta_j, \gamma_j, \varphi_1)$ where $u_j \in V(T_0)$ with $\gamma_j \in \overline{\varphi}_1(u_j)$ and $P'_j = P_{u_j}(\delta_j, \gamma_j, \varphi_2)$. Since T_{j+1} is (φ_1, φ_2)-stable and $f_j, f_j^s \in E(T_{j+1})$, it follows $\varphi_2(f_j) = \delta_j$ and $\varphi_2(f_j^s) = \gamma_j$, and so f_j is the first edge of $E_{\delta_j}^{\varphi_2}(\partial(T_j))$ contained in P'_j and f_j^s is

the next edge of f_j in P'_j. Again, since T_{j+1} is (φ_1, φ_2)-stable, (T_{j+1}, φ_1) is an ETT and $T_{j+1} \subsetneq T$, it follows that P'_j contains $E^{\varphi_2}_{\delta_j}(\partial(T_j))$ by the minimality of T. Therefore, (f_j, f^s_j) is a connecting edge pair of T_j in φ_2. Thus, (T_m, φ_2) is an ETT equivalent to (T_m, φ_1). We point out that $P_{u_j}(\gamma_j, \delta_j, \varphi_1)$ and $P_{u_j}(\gamma_j, \delta_j, \varphi_2)$ may be different.

Thus (2) holds and there exist $\gamma \in \Gamma^f(T_0) - \varphi_2(E(T_m))$ and $\delta \notin \overline{\varphi}_2(V(T_m))$ such that $P_u(\gamma, \delta, \varphi_2) \cap E^{\varphi_2}_\delta(\partial(T_m)) \neq \emptyset$ and $P_u(\gamma, \delta, \varphi_2) \not\supseteq E^{\varphi_2}_\delta(\partial(T_m))$. Let $P = P_u(\gamma, \delta, \varphi_2)$. Let Q be another (γ, δ)-chain such that $E(Q) \cap E^{\varphi_2}_\delta(\partial(T_m)) \neq \emptyset$. By the minimality of T, P contains every edge of $E^{\varphi_2}_\delta(\partial(T_{m-1}))$, and so $E(T_{m-1}) \cap E(Q) = \emptyset$. Let $\varphi_3 := \varphi_2/Q$ be a coloring of $G - e$ obtained from φ_2 by interchanging colors assigned on $E(Q)$. Then, T_{m-1} is (φ_2, φ_3)-stable and so $\varphi_3 \in \Phi$.

Let (f, f^s) be the connecting edge pair of T_{m-1} in (T, φ_1), and $T' = T_{m-1} + (f, f^s)$. We claim that $E(T') \cap E(Q) = \emptyset$. Since Q is disjoint with T_{m-1}, $f \notin E(Q)$. Since $\varphi_2(f^s) \in \overline{\varphi}_2(T_0)$, it follows that $\varphi_2(f^s) \neq \delta$, and since $\gamma \notin \varphi_1(E(T_m)) = \varphi_2(E(T_m))$ and $\varphi_2(f^s) \in \varphi_2(E(T_m))$, we have $\varphi_2(f^s) \neq \gamma$. That is, $f^s \notin E(Q)$.

Observe that (T', φ_1) is an ETT with the ladder $T_0 \subset \cdots \subset T_{m-1} \subset T'$ and T' is (φ_1, φ_3)-stable. Moreover $|T'| \leq |T_m| < |T|$. Therefore, by the minimality of T, (T_{m-1}, φ_3) is an ETT equivalent to (T_{m-1}, φ_1), and because we do not use any edge of Q when we extend (T_{m-1}, φ_1) to (T_m, φ_1), we can extend (T_{m-1}, φ_3) to (T_m, φ_3) equivalent to (T_m, φ_1). However, T_m is not closed in φ_3, which yields a contradiction that (T, φ_1) is a maximal ETT. This completes the proof. □

We easily obtain the following two lemmas from Lemma 3.2.

Lemma 3.3. *Let (T, φ) be a maximal ETT with ladder $T_0 \subset T_1 \subset \cdots \subset T_m \subset T$. If T_m is elementary and $\Gamma^f(T_0) - \varphi(E(T)) \neq \emptyset$, then for any $\gamma \in \Gamma^f(T_0) - \varphi(E(T))$ and $\delta \notin \overline{\varphi}(V(T_m))$, $P_u(\gamma, \delta, \varphi) \supset E^\varphi_\delta(\partial(T_i))$ for every i with $0 \leq i \leq m$, where $u \in V(T_0)$ such that $\gamma \in \overline{\varphi}(u)$.*

Proof. Let $P := P_u(\gamma, \delta, \varphi)$. Since T is elementary, u is the unique vertex in $V(T)$ such that $\gamma \in \overline{\varphi}(u)$. So, the end of P, other than u, is not in $V(T)$, which gives that P meets $E^\varphi_\delta(\partial(T_i))$ for every $i = 0, 1, \ldots, m$. Therefore, Lemma 3.2 implies that $P \supset E^\varphi_\delta(\partial(T_i))$ for every i with $0 \leq i \leq m$. □

Lemma 3.4. *Let (T, φ) be a maximal ETT with largest closed segment (T_m, φ). Let (f, f^s) be the connecting edge pair of T_m in (T, φ). Let $\varphi' \in \Phi_T$ be a coloring such that T_m is (φ, φ')-stable. If $\varphi(f^s) = \varphi'(f^s)$ then (f, f^s) is a connecting edge pair of T_m in φ' and (Tf^s, φ') is an ETT equivalent to (Tf^s, φ).*

Proof. By Lemma 3.2, (T_m, φ') is an ETT equivalent to (T_m, φ). Let $\varphi(f) = \delta$, $\varphi(f^s) = \gamma$ and $u \in V(T_0)$ with $\gamma \in \overline{\varphi}(u)$. Let $P := P_u(\delta, \gamma, \varphi)$ and $P' := P_u(\delta, \gamma, \varphi')$. Since T_m is (φ, φ')-stable and $\varphi(f^s) = \varphi'(f^s)$, f is the first edge of $E^{\varphi'}_\delta(\partial(T_m))$ contained in P' and f^s is the next edge of f in P'. By Lemma 3.2, P' contains every edge of $E^{\varphi'}_\delta(\partial(T_m))$, so (f, f^s) is a connecting edge pair of T_m in φ'. This completes the proof. □

For two ETTs (T, φ) and (T', φ'), we say (T, φ) *contains* (T', φ') *as a premier segment*, if (T', φ) is a premier segment of (T, φ) and (T', φ') is equivalent to (T', φ). That is, T' is an initial subsequence of T, and for $f \in T'$, f is a connecting edge of (T', φ') if and only if f is a connecting edge of (T, φ).

Lemma 3.5. *For every ETT (T, φ), if T is elementary such that $|\Gamma^f(T_0)| > m(T, \varphi)$ and $|E(T) - E(T_0)| - m(T, \varphi) < |\overline{\varphi}(V(T_0))|$, then there exists an ETT (T^*, φ^*) containing (T, φ) as a premier segment.*

Proof. Let $m = m(T, \varphi)$ and (f_i, f_i^s) be connecting edge pair of (T, φ) for $i = 1, 2, \ldots, m$. Let $\varphi(f_i^s) = \gamma_i$.

Claim 3.1. *There exists $\varphi' \in \Phi$ such that (T, φ') is an ETT equivalent to (T, φ), T is elementary in φ' and there exists $\gamma' \in \overline{\varphi}'(V(T_0)) - \varphi'(E(T))$.*

Proof of Claim 3.1. Since $\varphi(f_i) \notin \overline{\varphi}(V(T_0))$ for each connecting edge f_i, we have $|\varphi(E(T) - E(T_0)) \cap \overline{\varphi}(V(T_0))| \leq |E(T) - E(T_0)| - m < |\overline{\varphi}(V(T_0))|$. So, $\overline{\varphi}(V(T_0)) - \varphi(E(T) - E(T_0)) \neq \emptyset$. Let $\gamma \in \overline{\varphi}(V(T_0)) - \varphi(E(T) - E(T_0))$.

We may assume that $\gamma \in \varphi(E(T_0))$ since otherwise we obtain a desired coloring $\varphi' = \varphi$ with $\gamma' = \gamma$. Since $m < |\Gamma^f(T_0)|$, there exists a color $\beta \in \Gamma^f(T_0) - \{\gamma_1, \gamma_2, \ldots, \gamma_m\}$. Since T_0 is closed, a (β, γ)-chain is either in $G[V(T_0)]$ or vertex disjoint with T_0. Let φ' be obtained from φ by interchanging β and γ for edges in $E_\beta^\varphi(G - V(T_0)) \cup E_\gamma^\varphi(G - V(T_0))$. Clearly, T_0 is (φ, φ')-stable. Since γ and β are missing colors of T_0 not used on any connecting edge pair, (T, φ') is an ETT equivalent to (T, φ), and since $\beta \in \Gamma^f(T_0)$ and $\gamma \notin \varphi(E(T) - E(T_0))$, we have $\beta \notin \varphi'(E(T))$. We also obtain $\overline{\varphi}(v) = \overline{\varphi}'(v)$ for every $v \in V(T)$ since $\beta, \gamma \notin \overline{\varphi}(v)$ for every $v \in V(T) - V(T_0)$. So, T is elementary in φ'. Therefore, we obtain a desired coloring φ' with $\gamma' = \beta$. $\square$

We consider $\varphi' \in \Phi$ and $\gamma' \in \overline{\varphi}'(V(T_0)) - \varphi'(E(T))$ described in Claim 3.1. In φ', if either T is not closed or has a connecting edge, then we can apply **ET0** and **ET1** to extend (T, φ') to a larger ETT (T', φ'). Suppose T is closed and does not have a connecting edge in φ'. In this case, T is both elementary and closed. Let the ladder of (T, φ') be $T_0 \subset T_1 \subset \cdots \subset T_m \subset T$. Since G itself is not elementary, T is not strongly closed. Thus, T has a defective color δ. Since T does not have a connecting edge, $P_v(\gamma', \delta, \varphi')$ does not contain all edges of $E_\delta^{\varphi'}(\partial(T))$, where $v \in V(T_0)$ is the unique vertex with $\gamma' \in \overline{\varphi}'(v)$. Let Q be another (γ', δ)-chain containing some edges in $E_\delta^{\varphi'}(\partial(T))$ and let $\varphi^* = \varphi/Q$. By Lemma 3.3, Q is disjoint with T_m. So, T_m is (φ', φ^*)-stable. By Lemma 3.2, (T_m, φ^*) is an ETT equivalent to (T_m, φ'). Since $\gamma', \delta \neq \varphi'(f_m^s)$ and Q is disjoint with $V(T_m)$, (T_m, φ^*) can be extended to (T, φ^*) as an ETT equivalent to (T, φ'). Now, since T is not closed in φ^*, by applying **ET1**, we can extend (T, φ^*) to a larger ETT (T^*, φ^*), which contains (T, φ) as a premier segment. This completes the proof. $\square$

3.2. The major result

The following theorem and its corollary show that if G is not elementary, there exists a large ETT (T, φ) such that T is elementary in φ. This is fundamental for both Theorems 1.1 and 1.2.

Theorem 3.1. *Let G be a k-critical graph with $k \geq \Delta + 1$ and (T, φ) be a maximal ETT. If $|E(T) - E(T_0)| - m(T, \varphi) < |\overline{\varphi}(V(T_0))| - 1$ and $m(T, \varphi) < |\Gamma^f(T_0)| - 1$, then T is elementary.*

Combining Theorem 3.1 and Lemma 3.5, we obtain the following result.

Corollary 3.1. *Let G be a k-critical graph with $k \geq \Delta + 1$. If G is not elementary, then there is an ETT (T, φ) based on $T_0 \in \mathcal{T}$ with m rungs such that T is elementary and*

$$|T| \geq |T_0| - 2 + \min\{m + |\overline{\varphi}(V(T_0))|, 2(|\Gamma^f(T_0)| - 1)\}.$$

Proof. Let (T, φ) be an ETT with maximum $|T|$ such that T is elementary in φ. Let (T, φ) have m rungs. We may assume that $|T| < |T_0| - 2 + \min\{m + |\overline{\varphi}(V(T_0))|, 2(|\Gamma^f(T_0)| - 1)\}$, that is, $|T| < |T_0| - 2 + m + |\overline{\varphi}(V(T_0))|$ and $|T| < |T_0| - 4 + 2|\Gamma^f(T_0)|$. Thus,

$$|E(T) - E(T_0)| - m = |T| - |T_0| - m < |\overline{\varphi}(V(T_0))| - 2$$

and

$$m \leq \frac{|T| - |T_0|}{2} < |\Gamma^f(T_0)| - 2,$$

which give that there exists an ETT (T^*, φ^*) containing (T, φ) as a premier segment by Lemma 3.5, where either $|T^* - T| = 1$ or $T^* = T + (f, f^s)$ and (f, f^s) is a connecting edge pair of T in (T^*, φ^*). In both cases, we have $|E(T^*) - E(T_0)| - m(T^*, \varphi^*) < |\overline{\varphi}^*(V(T_0))| - 1$ and $m(T^*, \varphi^*) < |\Gamma^f(T_0)| - 1$. By Theorem 3.1, T^* is elementary in φ^*, a contradiction to the maximality of T. This completes the proof. $\square$

The remaining of this section will be devoted to proving Theorem 3.1.

To prove Theorem 3.1, we consider a minimal counterexample to Theorem 3.1 and investigate its properties in order to reach a contradiction. Let T be a minimum sequence of edges of G such that there exists $\varphi \in \Phi$ where (T, φ) is a counterexample to Theorem 3.1, that is, $|E(T) - E(T_0)| - m(T, \varphi) < |\overline{\varphi}(V(T_0))| - 1$ and $m(T, \varphi) < |\Gamma^f(T_0)| - 1$, but T is non-elementary.

Let $\Phi_T \subseteq \Phi$ be the set of all colorings φ such that (T, φ) is a counterexample to Theorem 3.1. For $\varphi \in \Phi_T$, let $L(\varphi) = \{\varphi(f^s) \mid f^s$ is the companion of a connecting edge of $(T, \varphi)\}$.

For $\varphi \in \Phi_T$, for any premier segment (T', φ) of (T, φ), (T', φ) is maximal by Lemma 3.1. Additionally, following the definition, we can verify that $|E(T') - E(T_0)| - m(T', \varphi) \leq |E(T) - E(T_0)| - m(T, \varphi)$ and $m(T', \varphi) \leq m(T, \varphi)$. So, every premier segment of (T, φ) satisfies the conditions of Theorem 3.1 and is elementary.

For $\varphi \in \Phi_T$ and a color α, we say α is a *common missing color of* φ if there exist $v_1, v_2 \in V(T)$ such that $\alpha \in \overline{\varphi}(v_1) \cap \overline{\varphi}(v_2)$. For $\varphi \in \Phi_T$ and its missing color α, let $\Psi(\varphi, \alpha) = \overline{\varphi}(V(T_0)) \cap (\varphi(E(T) - E(T_0)) \cup \{\alpha\})$, that is, the set of all missing colors of T_0 used on $T - T_0$ or equal to α. In the following series of claims, we will prove that for $\varphi \in \Phi_T$, we can always construct $\varphi' \in \Phi_T$ with $|\Psi(\varphi', \alpha')| > |\Psi(\varphi, \alpha)|$ where α' and α are common missing colors of φ' and φ, respectively, which gives a contradiction because $|\Psi(\varphi, \alpha)|$ can not be larger than $|\overline{\varphi}(V(T_0))|$.

A free color which is not used in $E(T)$ plays a crucial role to reach a contradiction. The following claim shows that for a given $\varphi \in \Phi_T$, there exists $\varphi' \in \Phi_T$, which is not very different from φ such that $\Psi(\varphi, \alpha)$ contains a free color not used in $E(T)$.

Claim 3.2. *Let $\varphi \in \Phi_T$ and α be a common missing color of φ. Then, there exists $\varphi' \in \Phi_T$ such that*

(1) $(T, \varphi) \sim (T, \varphi')$,
(2) *for $f \in E(T)$, $\varphi(f) = \alpha$ if and only if $\varphi'(f) = \alpha$, and for $v \in V(T)$, $\alpha \in \overline{\varphi}(v)$ if and only if $\alpha \in \overline{\varphi}'(v)$, so α is a common missing color of φ',*
(3) *there exists $\gamma \in \Gamma^f(T_0) - (\varphi'(E(T)) \cup \{\alpha\})$, and*
(4) $|\Psi(\varphi, \alpha)| = |\Psi(\varphi', \alpha)|$.

Recall that $|E(T) - E(T_0)| - m(T, \varphi) < |\overline{\varphi}(V(T_0))| - 1$ and $m(T, \varphi) < |\Gamma^f(T_0)| - 1$ for every $\varphi \in \Phi_T$.

Proof of Claim 3.2. Let $X = \Gamma^f(T_0) - (L(\varphi) \cup \{\alpha\})$ and $Y = \overline{\varphi}(V(T_0)) - \big(\varphi(E(T) - E(T_0)) \cup \{\alpha\}\big)$. Since $|L(\varphi)| = m(T, \varphi) < |\Gamma^f(T_0)| - 1$, we have $X = \Gamma^f(T_0) - (L(\varphi) \cup \{\alpha\}) \neq \emptyset$. In addition, since $|\varphi(E(T) - E(T_0))| \leq |E(T) - E(T_0)| - m(T, \varphi) < |\overline{\varphi}(V(T_0))| - 1$, we have $Y = \overline{\varphi}(V(T_0)) - \big(\varphi(E(T) - E(T_0)) \cup \{\alpha\}\big) \neq \emptyset$.

Let $\omega \in X$. If ω is not used in $E(T) - E(T_0)$, then $\omega \in \Gamma^f(T_0) - (L(\varphi) \cup \{\alpha\}) - \varphi(E(T) - E(T_0)) = \Gamma^f(T_0) - (\varphi(E(T)) \cup \{\alpha\})$. Then, φ, with $\gamma = \omega$, satisfies the conditions. So, we may assume that $\omega \in \big(\Gamma^f(T_0) - (L(\varphi) \cup \{\alpha\})\big) \cap \varphi(E(T) - E(T_0))$. Let $\beta \in Y$. If $\beta \in \Gamma^f(T_0)$, then $\beta \in \Gamma^f(T_0) - \big(\varphi(E(T) - E(T_0)) \cup \{\alpha\}\big) = \Gamma^f(T_0) - (\varphi(E(T)) \cup \{\alpha\})$, so φ, with $\gamma = \beta$, satisfies the conditions. Hence, we may assume that $\beta \in (\overline{\varphi}(V(T_0)) - \Gamma^f(T_0)) - \big(\varphi(E(T) - E(T_0)) \cup \{\alpha\}\big)$.

Since T_0 is closed in φ and $\beta, \omega \in \overline{\varphi}(V(T_0))$, a (β, ω)-chain is either in $G[V(T_0)]$ or disjoint with $V(T_0)$. Let φ_1 be obtained from φ by interchanging colors β and ω on all (β, ω)-chains disjoint with $V(T_0)$. We claim that φ_1 is a desired coloring. Clearly, (2) holds since $\alpha \neq \beta, \omega$. Since T_0 is (φ, φ_1)-stable and both ω and β are in $\overline{\varphi}(V(T_0)) - L(\varphi)$, (T_0, φ_1) can be extended to (T, φ_1) as an ETT equivalent

to (T, φ). Since $\omega \in \Gamma^f(T_0) - \{\alpha\}$ and $\beta \notin \varphi(E(T) - E(T_0)) - \{\alpha\}$, it follows that $\omega \in \Gamma^f(T_0) - (\varphi_1(E(T)) \cup \{\alpha\})$. Finally, $|\Psi(\varphi, \alpha)| = |\Psi(\varphi', \alpha)|$ since $\beta \notin \Psi(\varphi, \alpha)$, $\omega \in \Psi(\varphi, \alpha)$, $\beta \in \Psi(\varphi', \alpha)$ and $\omega \notin \Psi(\varphi', \alpha)$. Therefore, φ_1, with $\gamma = \omega$, satisfies the conditions. This completes the proof. □

Claim 3.3. *For every $\varphi \in \Phi_T$, the last operation in adding the last edge of (T, φ) is* **ET1**.

Proof. Let Φ'_T be the set of all colorings $\varphi \in \Phi_T$ such that the last operation in adding the last edge of (T, φ) is **ET0**. Suppose Φ'_T is not empty. Let (f, f^s) be the last two edges of T. Clearly, for $\varphi \in \Phi'_T$, (f, f^s) is a connecting edge pair. Let x be the in-end of f, y be the out-end of f (in-end of f^s), and z be the out-end of f^s. Let $T' = T \setminus \{f, f^s\}$. Note that $V(T')$ is closed in φ for every $\varphi \in \Phi'_T$.

(1) For every $\varphi \in \Phi'_T$, $\overline{\varphi}(y) \cap \overline{\varphi}(z) = \emptyset$ and $\overline{\varphi}(y) \cap \overline{\varphi}(V(T')) = \emptyset$.

Let $\varphi(f) = \delta$ and $\varphi(f^s) = \gamma_\delta \in \Gamma^f(T_0) - \varphi(E(T'))$. Note that f is the first edge in $\partial(E(T'))$ along $P := P_u(\gamma_\delta, \delta, \varphi)$ with color δ, where $u \in V(T_0)$ such that $\gamma_\delta \in \overline{\varphi}(u)$. In addition, $|E^\varphi_\delta(\partial(T'))| \geq 2$ since δ is a defective color.

Suppose $\alpha \in \overline{\varphi}(y) \cap \overline{\varphi}(z)$ for some color α. Then, f^s can be recolored with α and let this new coloring be φ_1. Let $P' = P_u(\gamma_\delta, \delta, \varphi_1)$. Clearly, $\gamma_\delta \in \Gamma^f(T_0) - \varphi_1(V(T'))$ and $\delta \notin \overline{\varphi}_1(V(T'))$. Since T' is (φ, φ_1)-stable, P' contains $E^{\varphi_1}_\delta(\partial(T'))$ by Lemma 3.2. However, P' is the subpath of P ending in y, so $E(P') \cap E^{\varphi_1}_\delta(\partial(T')) = \{f\}$, which yields a contradiction since $|E^{\varphi_1}_\delta(\partial(T'))| = |E^\varphi_\delta(\partial(T'))| \geq 2$. Therefore, $\overline{\varphi}(y) \cap \overline{\varphi}(z) = \emptyset$.

Suppose $\alpha \in \overline{\varphi}(y) \cap \overline{\varphi}(V(T'))$ for some color α. Since $\delta, \gamma_\delta \in \varphi(y)$ and $\alpha \in \overline{\varphi}(y)$, we have $\alpha \notin \{\delta, \gamma_\delta\}$. Consider the (α, γ_δ)-chain $P'' := P_y(\alpha, \gamma_\delta, \varphi)$. Since $\alpha, \gamma_\delta \in \overline{\varphi}(V(T'))$ and T' is closed, $V(P'') \cap V(T_m) = \emptyset$. Let $\varphi_1 = \varphi/P''$. Clearly, T' is (φ, φ_1)-stable, which gives that $E(P_u(\gamma_\delta, \delta, \varphi_1)) \supseteq E^{\varphi_1}_\delta(\partial(T_m))$ by Lemma 3.2. However, $P_u(\gamma_\delta, \delta, \varphi_1)$ is the subpath of $P_u(\gamma_\delta, \delta, \varphi)$ from u to y, which is a contradiction. Therefore, $\overline{\varphi}(y) \cap \overline{\varphi}(V(T')) = \emptyset$. This proves (1).

The minimality of T and (1) imply that for every $\varphi \in \Phi'_T$, every common missing color α of φ is contained in $\overline{\varphi}(z) \cap \overline{\varphi}(V(T'))$.

(2) There exists $\varphi \in \Phi'_T$ such that $\alpha \in \Gamma^f(T_0) - \varphi(E(T))$ for some common missing color α of φ.

Since Φ'_T is not empty, Claim 3.2 shows that there exists $\varphi \in \Phi'_T$ with common missing color α such that there exists $\gamma \in \Gamma^f(T_0) - (\varphi(E(T)) \cup \{\alpha\})$. Clearly, $\alpha \in \overline{\varphi}(z) \cap \overline{\varphi}(V(T'))$. Consider the (α, γ)-chain $P_1 := P_z(\alpha, \gamma, \varphi)$. Since $\alpha, \gamma \in \overline{\varphi}(V(T'))$ and $V(T')$ is closed with respect to φ, we have $V(P_1) \cap V(T') = \emptyset$. Let $\varphi_1 = \varphi/P_1$. Since T' is (φ, φ_1)-stable and $\alpha, \gamma \neq \varphi(f^s)$, it follows that (f, f^s) is a connecting edge pair of T' in φ_1 and (T, φ_1) is an ETT equivalent to (T, φ) by Lemma 3.4. Furthermore, γ is a common missing color of φ_1, so $\varphi_1 \in \Phi'_T$. Finally, since $\gamma \in \Gamma^f(T_0) - \varphi_1(E(T))$, we obtain a desired coloring φ_1 and its common missing color γ. This proves (2).

We consider $\varphi \in \Phi'_T$ and its common missing color α described in (2), that is, $\alpha \in \Gamma^f(T_0) - \varphi(E(T))$. Let $\varphi(f) = \delta$ and $\varphi(f^s) = \gamma_\delta$. Recall that $\alpha \in \overline{\varphi}(z) \cap \overline{\varphi}(V(T'))$ which gives $\alpha \notin \{\delta, \gamma_\delta\}$. Let v be the unique vertex in $V(T_0)$ with $\alpha \in \overline{\varphi}(v)$. Let $\beta \in \overline{\varphi}(y)$. Note

that $\beta \notin \overline{\varphi}(V(T'))$ by (1). Let $P_v := P_v(\alpha, \beta, \varphi)$, $P_y := P_y(\alpha, \beta, \varphi)$ and $P_z := P_z(\alpha, \beta, \varphi)$. We claim that $P_v = P_y$. Suppose, on the contrary, that $P_v \neq P_y$. Since $\alpha \in \Gamma^f(T_0)$ and $\beta \notin \overline{\varphi}(V(T'))$, $E(P_v) \supset E^{\varphi}_{\beta}(\partial(T'))$ by Lemma 3.3. Therefore, $V(P_y) \cap V(T') = \emptyset$. Let $\varphi_1 = \varphi/P_y$. Then, since T' is (φ, φ_1)-stable and $\alpha, \beta \neq \gamma_\delta$, (f, f^s) is a connecting edge pair of T' in φ_1 and (T, φ_1) is an ETT equivalent to (T, φ) by Lemma 3.4. However, (T, φ_1) has a common missing color $\alpha \in \overline{\varphi}_1(y) \cap \overline{\varphi}_1(V(T'))$ which yields a contradiction to (1). Hence, $P_v = P_y$ and it is vertex disjoint with P_z. Let $\varphi_2 = \varphi/P_z$. By Lemma 3.3, $E(P_v) \supset E^{\varphi}_{\beta}(\partial(T'))$. So, $V(P_z) \cap V(T') = \emptyset$, which in turn gives that T' is (φ, φ_2)-stable. Since $\alpha, \beta \neq \gamma_\delta$, (f, f^s) is a connecting edge pair of T' in φ_2 by Lemma 3.4 and (T, φ_2) is an ETT, equivalent to (T, φ), with a common missing color $\beta \in \overline{\varphi}_2(y) \cap \overline{\varphi}_2(z)$, which is a contradiction to (1). Therefore, Φ'_T is empty and the last operation in (T, φ) is **ET1** for every $\varphi \in \Phi_T$. $\square$

By Claim 3.3, the last edge is added to (T, φ) by **ET1** for every $\varphi \in \Phi_T$. Let y and z be the in-end and out-end of f, respectively, and let $T' = T - f$. By the minimality of T, every common missing color of φ is in $\overline{\varphi}(z) \cap \overline{\varphi}(V(T'))$.

Claim 3.4. *Let (T_m, φ) be the largest closed segment of (T, φ) and (T'', φ) be a premier segment of (T, φ) with $T_m \subset T'' \neq T$. For any color $\gamma \in \Gamma^f(T_0) - \varphi(E(T''))$ and any color $\beta \in \overline{\varphi}(V(T''))$, let $u \in V(T_0)$ such that $\gamma \in \overline{\varphi}(u)$ and $v \in V(T'')$ such that $\beta \in \overline{\varphi}(v)$. Then u and v are on the same (β, γ)-chain.*

Proof. Since T_m is both elementary and closed in φ, u and v are on the same (β, γ)-chain if $v \in V(T_m)$. Suppose $v \in V(T'') - V(T_m)$ and, on the contrary, $P_u := P_u(\gamma, \beta, \varphi)$ and $P_v := P_v(\gamma, \beta, \varphi)$ are vertex disjoint. By Lemma 3.3, $E(P_u) \supset E^{\varphi}_{\beta}(\partial(T_m))$, so $V(P_v) \cap V(T_m) = \emptyset$. Let $\varphi_1 = \varphi/P_v$ be the coloring obtained by interchanging the colors β and γ on $P_v(\gamma, \beta, \varphi)$. Clearly, T_m is (φ, φ_1)-stable. Let (f, f^s) be the connecting edge pair of T_m in (T'', φ). Observe that $\gamma, \delta \neq \varphi(f^s)$. Therefore, $\varphi(f^s) = \varphi_1(f^s)$, which gives that (Tf^s, φ_1) is an ETT by Lemma 3.4. If v is the intersection of f and f^s, we reach a contradiction since (Tf^s, φ_1) is a counterexample to Theorem 3.1 with $|Tf^s| < |T|$. So, we may assume v is not the intersection of f and f^s. As $\gamma \notin \varphi(E(T''))$, we can extend (Tf^s, φ_1) to (T'', φ_1). But, in the coloring φ_1, $\gamma \in \overline{\varphi}_1(u) \cap \overline{\varphi}_1(v)$, which gives a contradiction to the minimality of $|T|$. $\square$

So far, we proved all claims for a fixed minimal counterexample T. However, indeed the claims hold for every minimal counterexample to Theorem 3.1. In particular, Claim 3.4 is true for every minimal counterexample. We will use it later.

Claim 3.5. *Let $\varphi \in \Phi_T$ and (T_m, φ) the largest closed segment of (T, φ). Let α be a common missing color of φ. Then, there exist $\varphi' \in \Phi_T$ and its common missing color α' such that*

- $(T, \varphi') \sim (T, \varphi)$,

- $\alpha' \in \Gamma^f(T_0) - \varphi'(E(T_m))$,
- $|\Psi(\varphi', \alpha')| \geq |\Psi(\varphi, \alpha)|$.

Proof. Let (f_m, f_m^s) be the connecting edge pair of T_m in (T, φ).
(1) There exist $\varphi_1 \in \Phi_T$ and its common missing color α_1 such that (T, φ_1) is equivalent to (T, φ), $\alpha_1 \in \overline{\varphi}_1(V(T_m))$ and $|\Psi(\varphi, \alpha)| \leq |\Psi(\varphi_1, \alpha_1)|$.

We may assume that $\alpha \notin \overline{\varphi}(V(T_m))$, that is, $\alpha \in \overline{\varphi}(V(T') - V(T_m))$. By Claim 3.2, we may assume that there exists $\gamma \in \Gamma^f(T_0) - \big(\varphi(E(T)) \cup \{\alpha\}\big)$. Let u be the vertex of $V(T_0)$ such that $\gamma \in \overline{\varphi}(u)$ and v the vertex of $V(T') - V(T_m)$ such that $\alpha \in \overline{\varphi}(v)$. By Claim 3.4, $P_u(\alpha, \gamma, \varphi) = P_v(\alpha, \gamma, \varphi)$ and is disjoint with $P = P_z(\alpha, \gamma, \varphi)$. Since $\alpha \notin \overline{\varphi}(V(T_m))$, $P_u(\alpha, \gamma, \varphi)$ contains $E_\alpha^\varphi(\partial(T_m))$, so P is disjoint with $V(T_m)$. Let $\varphi_1 = \varphi/P$. Then, T_m is (φ, φ_1)-stable. Since $\alpha \notin \overline{\varphi}(V(T_m))$ and $\gamma \notin \varphi(E(T))$, we have $\alpha, \gamma \neq \varphi(f_m^s)$, which gives that (f_m, f_m^s) is a connecting edge pair of T_m in φ_1 by Lemma 3.4. Moreover, since $\gamma \notin \varphi(T)$, (Tf_m^s, φ_1) can be extended to (T, φ_1) equivalent to (T, φ). Note that γ is a common missing color of φ_1 since $\gamma \in \overline{\varphi}_1(z) \cap \overline{\varphi}_1(V(T_m))$. Finally, since $\alpha, \gamma \notin \Psi(\varphi, \alpha)$ and $\gamma \in \Psi(\varphi_1, \gamma)$, we have $|\Psi(\varphi_1, \gamma)| \geq |\Psi(\varphi, \alpha)|$. Therefore, φ_1, with $\alpha_1 = \gamma$, satisfies the conditions. This proves (1).

We consider φ_1 and a common missing color α_1 obtained from (1). So, $\alpha_1 \in \overline{\varphi}_1(T_m)$. Now we prove that there exist $\varphi' \in \Phi_T$ and a common missing color α' of φ' such that $(T, \varphi_1) \sim (T, \varphi')$, $\alpha' \in \Gamma^f(T_0) - \varphi'(E(T_m))$ and $|\Psi(\varphi', \alpha')| \geq |\Psi(\varphi_1, \alpha_1)|$, which proves Claim 3.5.

By applying Claim 3.2 to φ_1, we obtain φ_1' satisfying conditions (1), (2), (3) and (4) of Claim 3.2. By (2) of Claim 3.2, α_1 is a common missing color of φ_1' contained in $\overline{\varphi}_1'(V(T_m))$. If $\alpha_1 \in \Gamma^f(T_0) - \varphi_1'(E(T_m))$, then we are done. So, we may assume that $\alpha_1 \notin \Gamma^f(T_0) - \varphi_1'(E(T_m))$. Let $\gamma_1 \in \Gamma^f(T_0) - \big(\varphi_1'(E(T)) \cup \{\alpha_1\}\big)$. Since $\alpha_1, \gamma_1 \in \overline{\varphi}_1(V(T_m))$ and T_m is closed in φ_1', it follows that $\alpha_1, \gamma_1 \notin \varphi_1'(\partial(T_m))$. Let φ' be obtained from φ_1' by interchanging colors α_1 and γ_1 for edges in $E_{\alpha_1}(G - V(T_m)) \cup E_{\gamma_1}(G - V(T_m))$. Note that φ' is a proper edge coloring. Clearly, T_m is (φ_1', φ')-stable. Since $\alpha_1, \gamma_1 \notin \Gamma^f(T_0) - \varphi_1'(E(T_m))$, we have $\varphi_1'(f_m^s) \neq \alpha_1, \gamma_1$, so (Tf_m^s, φ') is an ETT equivalent to (Tf_m^s, φ_1') by Lemma 3.4, and it can be extended to (T, φ') equivalent to (T, φ_1'). In the coloring φ', γ_1 is a common missing color contained in $\overline{\varphi}'(z)$ and $\Gamma^f(T_0) - \varphi'(E(T_m))$. In addition, $\Psi(\varphi', \gamma_1) - \Psi(\varphi_1', \alpha_1) = \{\gamma_1\}$ and $\Psi(\varphi_1', \alpha_1) - \Psi(\varphi', \gamma_1) \subseteq \{\alpha_1\}$, so $|\Psi(\varphi', \gamma_1)| \geq |\Psi(\varphi_1', \alpha_1)| = |\Psi(\varphi_1, \alpha_1)|$. Thus, φ', with $\alpha' = \gamma_1$, satisfies the conditions. This completes the proof. □

Let $\mathcal{A}$ be the set of all pairs (φ, α) of a coloring $\varphi \in \Phi_T$ and its common missing color α such that $\alpha \in \Gamma^f(T_0) - \varphi(E(T_m))$, where (T_m, φ) is the largest closed segment of (T, φ). Claim 3.5 shows that $\mathcal{A}$ is not empty.

Claim 3.6. *For every $(\varphi, \alpha) \in \mathcal{A}$, α is used in $E(T - T_m)$, i.e., $\alpha \in \varphi(E(T - T_m))$ where (T_m, φ) is the largest closed segment of (T, φ).*

Proof. Suppose on the contrary $\alpha \notin \varphi(E(T - T_m))$. So, $\alpha \in \Gamma^f(T_0) - \varphi(E(T))$. Recall that f is the last edge of T, and $T' = T - f$. Let $\varphi(f) = \theta$. We consider two cases:

$y \in V(T_m)$ and $y \in V(T') - V(T_m)$. Let $v \in V(T_0)$ be the vertex with $\alpha \in \overline{\varphi}(v)$. Let (f_m, f_m^s) be the connecting edge pair of T_m in (T, φ) and (δ_m, γ_m) its corresponding colors. Note that $\alpha \neq \delta_m$ since $\alpha \in \overline{\varphi}(V(T_m))$.

Case 1: $y \in V(T_m)$.

As $f \in \partial(T_m)$ is not a connecting edge and T_m is closed in φ, we know that there exists $w \in V(T) - V(T_m)$ such that $\theta \in \overline{\varphi}(w)$. Consider the (α, θ)-chain $P_v := P_v(\alpha, \theta, \varphi)$. By Lemma 3.3, $E(P_v) \supset E_\theta^\varphi(\partial(T_m))$. So, $f \in E(P_v)$ and z is the other end of P_v. Then, $P_w := P_w(\alpha, \theta, \varphi)$ is disjoint with P_v, which in turn shows $V(P_w) \cap V(T_m) = \emptyset$. Let $\varphi_1 = \varphi/P_w$. Since $V(P_w) \cap V(T_m) = \emptyset$, T_m is (φ, φ_1)-stable. Since $\alpha, \theta \neq \gamma_m$, (f_m, f_m^s) is a connecting edge pair of T_m in φ_1 and (Tf_m^s, φ_1) is an ETT equivalent to (Tf_m^s, φ) by Lemma 3.4. In addition, since α is not used in $\varphi(T - T_m)$, (Tf_m^s, φ_1) can be extended to (T', φ_1) as an ETT. Note that $\alpha \in \overline{\varphi}_1(v) \cap \overline{\varphi}_1(w)$. So, T' is not elementary, which gives a contradiction to the minimality of $|T|$.

Case 2: $y \in V(T') - V(T_m)$.

Let $\beta \in \overline{\varphi}(y)$ be a missing color of y. By the minimality of T, $\beta \notin \overline{\varphi}(V(T_m))$. We consider the following two cases according to whether y is the last vertex of $T' = T - z$.

Subcase 2.1: Let y be the last vertex of T' or $T' = T_m + (f_m, f_m^s)$. Let $P_v := P_v(\alpha, \beta, \varphi)$, $P_y := P_y(\alpha, \beta, \varphi)$ and $P_z := P_z(\alpha, \beta, \varphi)$ be (α, β)-chains containing vertices v, y and z, respectively. By Claim 3.4, we have $P_v = P_y$, so it is disjoint with P_z. By Lemma 3.3, $E(P_v) \supset E_\beta^\varphi(\partial(T_m))$, so $V(P_z) \cap V(T_m) = \emptyset$. Let $\varphi_1 = \varphi/P_z$ be the coloring obtained from φ by interchanging colors α and β on P_z. Clearly, T_m is (φ, φ_1)-stable. By Lemma 3.2, (T_m, φ_1) is an ETT. Since $\alpha \notin \varphi(E(T - T_m))$ and $\beta \notin \overline{\varphi}(V(T_m))$, we have $\alpha, \beta \neq \gamma_m$. So, (f_m, f_m^s) is a connecting edge pair of T_m in φ_1 and (Tf_m^s, φ_1) is an ETT equivalent to (Tf_m^s, φ) by Lemma 3.4. In addition, since α is not used in $T' - T_m$ in φ, (Tf_m^s, φ_1) can be extended to (T, φ_1). In the coloring φ_1, $\theta = \varphi_1(f)$ and f itself is a (β, θ)-chain. Let $\varphi_2 = \varphi_1/f$ be the coloring obtained from φ_1 by changing color θ to β on f. Since f is disjoint with T_m, we can verify that (T', φ_2) is an ETT by applying Lemma 3.2. Since f is not a connecting edge, $\theta \in \overline{\varphi}_2(V(T') - y) \cap \overline{\varphi}_2(y)$, which in turn shows that T' is not elementary with respect to φ_2, giving a contradiction to the minimality of T.

Subcase 2.2: Let y be not the last vertex of T' and $T' \neq T_m + (f_m, f_m^s)$. Let x be the last one, and f' be the edge of T' with out-end x. Observe that f' is added by **ET1** since $T' \neq T_m + (f_m, f_m^s)$, so $(T \setminus f', \varphi)$ is an ETT. If $\theta \in \varphi(x)$ then $(T - f', \varphi)$ is an ETT but $T - f'$ is not elementary, which contradicts the minimality of T. Hence $\theta \in \overline{\varphi}(x)$ and $\theta \notin \overline{\varphi}(V(T) - x)$.

We apply Claim 3.2 to φ and α. By (2) of Claim 3.2, instead of constructing a new coloring from Claim 3.2, we may assume that (3) of Claim 3.2 holds for φ and α. Let $\gamma \in \Gamma^f(T_0) - (\varphi(E(T)) \cup \{\alpha\})$, and $v_1 \in V(T_0)$ with $\gamma \in \overline{\varphi}(v_1)$.

Clearly $\alpha \in \varphi(x)$ by the minimality of T. Let $P_v := P_v(\alpha, \theta, \varphi)$, $P_x := P_x(\alpha, \theta, \varphi)$ and $P_z := P_z(\alpha, \theta, \varphi)$ be (α, θ)-chains containing vertices v, x and z, respectively. By Claim 3.4 we have $P_v = P_x$ which is disjoint with P_z. Lemma 3.3 implies that $E(P_v) \supset E_\theta^\varphi(\partial(T_m))$, and so we get $V(P_z) \cap V(T_m) = \emptyset$. Let $\varphi_1 = \varphi/P_z$ be the coloring

obtained from φ by interchanging colors α and θ along P_z. Observe that θ is only used on f for $E(T-(T_m \cup \partial(T_m)))$ since $\theta \in \overline{\varphi}(x)$, and f is colored by α in φ_1. Clearly T_m is (φ, φ_1)-stable. Note that $\theta \neq \gamma_m$ since x is the only vertex of $V(T')$ with $\theta \in \overline{\varphi}(x)$. So since $\alpha \neq \gamma_m$, (Tf_m^s, φ_1) is an ETT equivalent to (Tf_m^s, φ) by Lemma 3.4, and it can be extended to (T, φ_1) because $\alpha \notin \varphi(E(T))$. Observe that $\theta \in \overline{\varphi}_1(x) \cap \overline{\varphi}_1(z)$ and $\varphi_1 \in \Phi_T$. In addition, $\gamma \in \Gamma^f(T_0) - (\varphi_1(E(T)) \cup \{\theta\})$ because $\gamma \neq \alpha, \theta$. By Claim 3.4 the (γ, θ)-chain $P'_{v_1} := P_{v_1}(\gamma, \theta, \varphi_1)$ is the same with $P'_x := P_x(\gamma, \theta, \varphi_1)$, hence it is disjoint with $P'_z := P_z(\gamma, \theta, \varphi_1)$. Now we consider (T_{zx}, φ_2) obtained from (T, φ_2) by switching the order of adding edges f and f'. Clearly (T_{zx}, φ_2) is an ETT since f is colored by α in φ_2. Also, θ is a common missing color of φ_2, so (T_{zx}, φ_2) is another minimal counterexample to Theorem 3.1. Since Claim 3.4 works for every minimal counterexample, the (γ, θ)-chain $P'_{v_1} := P_{v_1}(\gamma, \theta, \varphi_2)$ is the same with $P'_z := P_z(\gamma, \theta, \varphi_2)$. Now we reach a contradiction. □

Claim 3.7. *For every $(\varphi, \alpha) \in \mathcal{A}$, there exists $\varphi' \in \Phi_T$ with common missing color α' such that $|\Psi(\varphi', \alpha')| > |\Psi(\varphi, \alpha)|$.*

Proof. Let (T_m, φ) be the largest closed segment of (T, φ). Let (f_m, f_m^s) be the connecting edge pair of T_m. By Claim 3.6, $\alpha \in E(T - T_m)$. Let $v \in V(T_0)$ be the vertex with $\alpha \in \overline{\varphi}(v)$. Following the linear order $\prec_T$, let e_1 be the first edge in $E(T - T_m)$ with $\varphi(e_1) = \alpha$, and let y_1 be the in-end of e_1. Pick a missing color $\beta_1 \in \overline{\varphi}(y_1)$. Note that, since $\varphi(e_1) = \alpha$ and $\alpha \in \Gamma^f(T_0) - \varphi(E(T_m))$, $e_1 \notin \partial(T_m)$. Hence $y_1 \in V(T) - V(T_m)$. Let $P_v := P_v(\alpha, \beta_1, \varphi)$, $P_{y_1} := P_{y_1}(\alpha, \beta_1, \varphi)$, and $P_z := P_z(\alpha, \beta_1, \varphi)$ be (α, β_1)-chains containing v, y_1 and z, respectively. By Claim 3.4, $P_v = P_{y_1}$, which in turn shows that it is disjoint with P_z. By Lemma 3.3, $E(P_v) \supset E^{\varphi}_{\beta_1}(\partial(T_m))$, so $V(P_z) \cap V(T_m) = \emptyset$.

Consider the coloring $\varphi_1 = \varphi/P_z$. Since $V(P_z) \cap V(T_m) = \emptyset$, T_m is (φ, φ_1)-stable, and (T_m, φ_1) is an ETT by Lemma 3.2. Clearly, $\alpha \neq \varphi(f_m^s)$ and since $\beta \notin \overline{\varphi}(T_0)$, $\beta \neq \varphi(f_m^s)$. Therefore, by Lemma 3.4, (Tf_m^s, φ_1) is an ETT. Since e_1 is the first edge colored with α along $\prec_T$, we have that $e_1 \prec_T e^*$ for all edges e^* colored with β_1. So, (T, φ_1) is an ETT. Note that $e_1 \in E(P_{y_1}) = E(P_v)$, which in turn gives $\varphi_1(e_1) = \alpha$. We also note that $\beta_1 \in \overline{\varphi}_1(z) \cap \overline{\varphi}_1(y_1)$, so $\varphi_1 \in \Phi_T$. Since $\beta_1 \notin \Psi(\varphi, \alpha) = \Psi(\varphi_1, \beta_1)$ and $\alpha \in \Psi(\varphi, \alpha) = \Psi(\varphi_1, \beta_1)$, it follows that $\Psi(\varphi, \alpha) = \Psi(\varphi_1, \beta_1)$.

Let φ_2 be a coloring obtained by applying Claim 3.2 to φ_1 and β_1. Note that $\beta_1 \notin \overline{\varphi}_1(T_m)$ since $\beta_1 \in \overline{\varphi}_1(y_1)$. Let $\gamma \in \Gamma^f(T_0) - \varphi_2(E(T))$ and $u \in V(T_0)$ such that $\gamma \in \overline{\varphi}_2(u)$. Let $Q_u := P_u(\gamma, \beta_1, \varphi_2)$, $Q_{y_1} := P_{y_1}(\gamma, \beta_1, \varphi_2)$ and $Q_z := P_z(\gamma, \beta_1, \varphi_2)$ be (γ, β_1)-chains containing u, y_1 and z, respectively. By Claim 3.4, $Q_u = Q_{y_1}$, so Q_u and Q_z are disjoint. By Lemma 3.3, $E(Q_u) \supset E^{\varphi_2}_{\beta_1}(\partial(T_m))$, so $V(Q_z) \cap V(T_m) = \emptyset$. Let $\varphi_3 = \varphi_2/Q_z$ be the coloring obtained from φ_2 by interchanging colors on Q_z. Since $V(Q_u) \cap V(T_m) = \emptyset$, (T_m, φ_3) is an ETT. Since $\gamma \notin \varphi_2(T)$ and β_1 is not a missing color of T_0, $\varphi_2(f_m^s) \neq \gamma, \beta$ and $\varphi_3(f_m^s) = \varphi_2(f_m^s)$. So, by Lemma 3.4, (T_m, φ_3) can be extended to (Tf_m^s, φ_3). In addition, (Tf_m^s, φ_3) can be extended to (T, φ_3) as an ETT. Since $\gamma \in \overline{\varphi}_3(z) \cap \overline{\varphi}_3(u)$, $\varphi_3 \in \Phi_T$ and γ is a common missing color of φ_3. Since $\beta_1 \notin \varphi_2(V(T_0)) = \varphi_3(V(T_0))$, $\gamma \in \Psi(\varphi_3, \gamma) - \Psi(\varphi_2, \beta_1)$, it follows that

$$|\Psi(\varphi_3, \gamma)| > |\Psi(\varphi_2, \beta_1)| = |\Psi(\varphi_1, \beta_1)| = |\Psi(\varphi, \alpha)|.$$

So, Claim 3.7 holds. □

Now we are ready to prove Theorem 3.1.

Proof of Theorem 3.1. Suppose there exists a counterexample. Following the definitions defined in this section, consider a minimal counterexample (T, φ) and its common missing color α such that $|\Psi(\varphi, \alpha)|$ is maximum. By Claim 3.5, there exist $\varphi_1 \in \Phi_T$ and a common missing color α_1 of φ_1 such that $(\varphi_1, \alpha_1) \in \mathcal{A}$ and $|\Psi(\varphi_1, \alpha_1)| \geq |\Psi(\varphi, \alpha)|$. Then, by Lemma 3.7, there exist $\varphi_2 \in \Phi_T$ and a common missing color α_2 of φ_2 such that $|\Psi(\varphi_2, \alpha_2)| > |\Psi(\varphi_1, \alpha_1)|$, which yields a contradiction that $|\Psi(\varphi, \alpha)|$ is maximum. This completes the proof. □

4. Proofs of Theorems 1.1 and 1.2

We start this section by outlining that how Corollary 3.1 is applied to show Theorems 1.1 and 1.2. Assuming G is not elementary, then G has a balanced Tashkinov tree T_0 of size at least $2(k - \Delta)$ by Corollary 2.2. Since each vertex of T_0 has at least $(k - \Delta)$ missing colors and T_0 is elementary, the total number of missing colors in T_0 is a quadratic function of $(k - \Delta)$, and so is the number of free colors of T_0. Applying Corollary 3.1, we can find an elementary ETT T based on T_0 such that its size is at least a quadratic function of $(k - \Delta)$. Since T is elementary and each vertex of T has at least $(k - \Delta)$ missing colors, we know that the total number of missing colors in T is a cubic function of $(k - \Delta)$. However, the total number of missing colors does not exceed k. This implies that k can only be about $\sqrt[3]{\Delta}$ away from Δ. For the proof of Theorem 1.2, based on the order of T_0 which is at least 11 guaranteed by Lemma 2.5, a similar argument shows that the order of T is at least 23. Thus any graph of order at most 23 is elementary. The part that any graph with $\Delta \leq 23$ is elementary is shown through only calculations.

4.1. Proof of Theorem 1.1

Clearly, we only need to prove Theorem 1.1 for critical graphs.

Theorem 4.1. *If G is a k-critical graph with $k \geq \Delta + \sqrt[3]{\Delta/2}$, then G is elementary.*

Proof. Suppose on the contrary that G is not elementary. Let $T_0 \in \mathcal{T}$ be a balanced Tashkinov tree. Lemma 2.3 shows that such a Tashkinov tree T_0 exists. Lemma 2.4 shows that $h(T_0) \geq 5$. By Corollary 3.1, there exists an ETT (T, φ) based on T_0 with m rungs such that $V(T)$ is elementary and

$$|T| \geq |T_0| - 2 + \min\{m + |\overline{\varphi}(V(T_0))|, 2(|\Gamma^f(T_0)| - 1)\}.$$

Since $m \geq 1$ and $|\overline{\varphi}(V(T_0))| \geq (k-\Delta)|T_0| + 2$, we have $|T_0| - 2 + m + |\overline{\varphi}(V(T_0))| \geq (k-\Delta+1)|T_0| + 1$. Since T_0 is a balanced Tashkinov tree, $|\varphi(E(T_0))| \leq \frac{|T_0|-1}{2}$, which in turn gives

$$|\Gamma^f(T_0)| = |\overline{\varphi}(V(T_0))| - |\varphi(E(T_0))| \geq (k-\Delta-\frac{1}{2})|T_0| + \frac{5}{2}.$$

Hence

$$|T_0| - 2 + 2(|\Gamma^f(T_0)| - 1) \geq 2(k-\Delta)|T_0| + 1 \geq (k-\Delta+1)|T_0| + 1.$$

Therefore, in any case, we have the following inequality

$$|T| \geq (k-\Delta+1)|T_0| + 1. \tag{1}$$

By Corollary 2.2, $|T_0| \geq 2(k-\Delta)+1$. Following (1), we get the inequality below.

$$|T| \geq (k-\Delta+1)(2(k-\Delta)+1)+1 = 2(k-\Delta)^2 + 3(k-\Delta) + 2. \tag{2}$$

Since T is elementary, we have $k \geq |\overline{\varphi}(V(T))| \geq (k-\Delta)|T| + 2$. Plugging into (2), we get the following inequality.

$$k \geq 2(k-\Delta)^3 + 3(k-\Delta)^2 + 2(k-\Delta) + 2.$$

Solving the above inequality, we obtain that $k < \Delta + \sqrt[3]{\Delta/2}$, giving a contradiction to $k \geq \Delta + \lceil \sqrt[3]{\Delta/2} \rceil$. □

4.2. Proofs of Theorem 1.2 and Corollary 1.1

We will need the following observation from [17]. For completeness, we give its proof here.

Lemma 4.1. *Let $s \geq 2$ be a positive integer and G be a k-critical graph with $k > \frac{s}{s-1}\Delta + \frac{s-3}{s-1}$. For any edge $e \in E(G)$, if $X \subseteq V(G)$ is an elementary set with respect to a coloring $\varphi \in \mathcal{C}^k(G-e)$ such that $V(e) \subseteq X$, then $|X| \leq s-1$.*

Proof. Otherwise, assume $|X| \geq s$. Since X is elementary, $k \geq |\overline{\varphi}(X)| \geq (k-\Delta)|X|+2 \geq s(k-\Delta)+2$, which in turn gives

$$\Delta \geq (s-1)(k-\Delta)+2 > (\Delta+(s-3))+2 = \Delta+s-1 > \Delta,$$

a contradiction. □

Clearly, to prove Theorem 1.2, it is sufficient to restrict our consideration to critical graphs.

Theorem 4.2. *If G is a k-critical graph with $k > \frac{23}{22}\Delta + \frac{20}{22}$, then G is elementary.*

Proof. Suppose, on the contrary, G is not elementary. Let $T_0 \in \mathcal{T}$ be a balanced Tashkinov tree with $h(T_0) \geq 5$. Lemma 2.3 and Lemma 2.4 show that such T_0 exists. By Corollary 3.1, there exists an ETT (T, φ) based on T_0 with m rungs such that $V(T)$ is elementary and

$$|T| \geq |T_0| - 2 + \min\{m + |\overline{\varphi}(V(T_0))|, 2(|\Gamma^f(T_0)| - 1)\}.$$

By Lemma 4.1, $|T| \leq 22$. We will show that $|T| \geq 23$ to lead a contradiction. By Lemma 2.5, we have $|T_0| \geq 11$. Since G is not elementary, $V(T_0)$ is not strongly closed, so $T \supsetneq T_0$. In particular, we have $m \geq 1$. Since $e \in E(T_0)$, we have $|\overline{\varphi}(V(T_0))| \geq |T_0| + 2$. Thus,

$$|T_0| - 2 + m + |\overline{\varphi}(V(T_0))| \geq 2|T_0| + 1 \geq 2 \times 11 + 1 = 23. \tag{3}$$

Since $h(T_0) \geq 5$, we obtain $|\varphi(E(T_0))| \leq (|T_0| - 1)/2$. So, $|\Gamma^f(T_0)| \geq |T_0| + 2 - (|T_0| - 1)/2 \geq (|T_0| + 5)/2$. Thus,

$$|T_0| - 2 + 2(|\Gamma^f(T_0)| - 1) \geq 2|T_0| + 1 \geq 23. \tag{4}$$

Combining (3) and (4), we get $|T| \geq 23$, giving a contradiction. □

We now give a proof of Corollary 1.1 and recall that Corollary 1.1 is stated as follows.

Corollary 4.1. *If G is a graph with $\Delta \leq 23$ or $|G| \leq 23$, then $\chi' \leq \max\{\Delta + 1, \lceil \chi'_f \rceil\}$.*

Proof. We assume that G is critical. Otherwise, we prove the corollary for a critical subgraph of G instead. If $\Delta \leq 23$, then $\lfloor \frac{23}{22}\Delta + \frac{20}{22} \rfloor = \lfloor \Delta + \frac{\Delta+20}{22} \rfloor \leq \Delta + 1$. If $\chi' \leq \Delta + 1$, we are done. Otherwise, we assume that $\chi' \geq \Delta + 2 \geq \frac{23}{22}\Delta + \frac{20}{22}$. By Theorem 1.2, we have $\chi' = \lceil \chi'_f \rceil$.

Assume that $|G| \leq 23$. If $\chi' \leq \Delta + 1$, then we are done. Otherwise, $\chi' = k + 1$ for some integer $k \geq \Delta + 1$. Let $T_0 \in \mathcal{T}$ be a balanced Tashkinov tree with $h(T_0) \geq 5$. By Corollary 3.1, there exists an ETT T based on $T_0 \in \mathcal{T}$ with m rungs such that $V(T)$ is elementary and

$$|T| \geq |T_0| - 2 + \min\{m + |\overline{\varphi}(V(T_0))|, 2(|\Gamma^f(T_0)| - 1)\}.$$

By Lemma 2.5, we have $|T_0| \geq 11$. Suppose that G is not elementary, then $V(T_0)$ is not strongly closed, so $T \supsetneq T_0$. In particular, we have $m \geq 1$. Since $e \in E(T_0)$, we have $|\overline{\varphi}(V(T_0))| \geq |T_0| + 2$. Thus,

$$|T_0| - 2 + m + |\overline{\varphi}(V(T_0))| \geq 2|T_0| + 1 \geq 2 \times 11 + 1 = 23. \tag{5}$$

Since $h(T_0) \geq 5$, we obtain $|\varphi(E(T_0))| \leq (|T_0| - 1)/2$. So, $|\Gamma^f(T_0)| \geq |T_0| + 2 - (|T_0| - 1)/2 \geq (|T_0| + 5)/2$. Thus,

$$|T_0| - 2 + 2(|\Gamma^f(T_0)| - 1) \geq 2|T_0| + 1 \geq 23. \tag{6}$$

Combining (5) and (6), we get $|T| \geq 23$. Then $|G| \geq |T| \geq 23$. Therefore, $|G| = 23$ and G is elementary, giving a contradiction. □

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